

A forecast that tells the baker how many loaves to make

Larkspur Bakehouse, a made up bakery with four shops, opens one page each morning and sees tomorrow's bake list, with the model and its test results underneath.

The problem

A bakery guesses tomorrow's bake from memory. Too few loaves loses sales, too many goes in the bin, and a chart of revenue does not tell anyone how many trays to put in the oven.

What I built

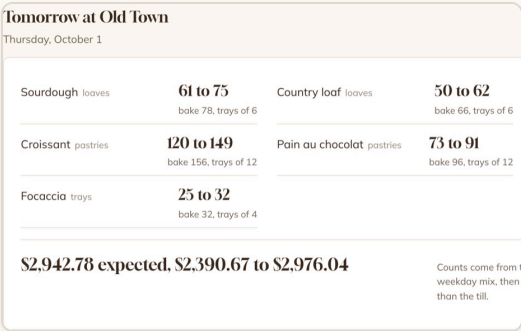
- A 28 day sales forecast per shop with an 80% range, tested on days the model never saw
- The forecast turned into a bake list per item, rounded up to whole trays
- A chart with real promotions and holidays marked, below the list
- A print view trimmed to the list, a CSV, and empty and error states
- Tests and the backtest rerun on every change by GitHub Actions

Measured results

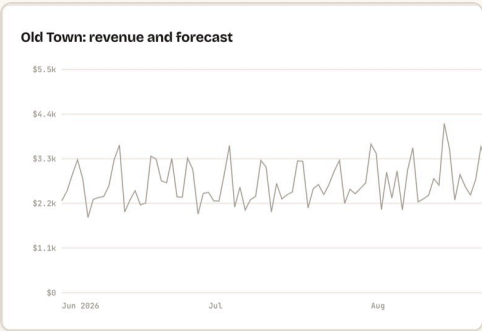
6.9% vs 11.3%
weighted error against simply repeating last week, over 672 unseen store days

79.2%
of days fell inside the 80% range

13 tests
including no leakage, the baseline, whole trays and closed days



Tomorrow's bake list



Revenue and the forecast



Model against the baseline

The model forecasts revenue, the number the till records. The bake list converts it to units with the shop's own prices and weekday mix, and the page says so.

Stack

Python

scikit-learn

pandas

SVG report

GitHub Actions

Live vanthienha199.github.io/larkspur-sales-forecast/
Code github.com/vanthienha199/larkspur-sales-forecast